

Algebraic Coding Theory

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Abstract

Notes taken while studying Algebraic Coding Theory, mostly from Hill book [1] and Lint book [2].

Usually while reading books and papers I take handwritten notes in a notebook, this document contains some of them re-written to *LaTeX*.

The proofs may slightly differ from the ones from the books, since I try to extend them for a deeper understanding.

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1 Vector spaces over finite fields, quick recap

Let q be a prime power.

$V(n, q)$: set $GF(q)^n$ of all ordered n -tuples over $GF(q)$. $V(n, q)$ is a vector space.

A set of vectors $\{v_1, v_2, \dots, v_r\}$ is said to be *linear dependent* if $\exists$ scalars $a_1, \dots, a_r$, not all zero, such that

$$a_1 v_1 + a_2 v_2 + \dots + a_r v_r = 0$$

They are *linear independent* if

$$a_1 v_1 + a_2 v_2 + \dots + a_r v_r = 0 \xRightarrow{\text{implies}} a_1 = a_2 = \dots = a_r = 0$$

Let C a subspace of $V(n, q)$. Then a subset $\{v_1, v_2, \dots, v_r\} \subseteq C$ is called a *generating set* (or *spanning set*) of C if every vector in C can be expressed as a linear combination of $v_1, \dots, v_r$.

A generating set of C which is also linearly independent is called a *basis* of C .

Theorem 4.3. Suppose $\{v_1, v_2, \dots, v_k\}$ is a basis of a subspace C of $V(n, q)$. Then

- i. every vector of C can be expressed uniquely as a linear combination of the basis vectors.
- ii. C contains exactly q^k vectors.

Proof. i. suppose vector $x \in C$ such that

$$x = a_1v_1 + a_2v_2 + \dots + a_kv_k$$

$$x = b_1v_1 + b_2v_2 + \dots + b_kv_k.$$

then

$$(a_1 - b_1)v_1 + (a_2 - b_2)v_2 + \dots + (a_k - b_k)v_k = 0.$$

But the set $\{v_1, v_2, \dots, v_k\}$ is linearly independent, and so

$$a_i - b_i = 0 \quad \text{for } i = 1, 2, \dots, k,$$

i.e.,

$$a_i = b_i \quad \text{for } i = 1, 2, \dots, k.$$

- ii. By (i), the q^k vectors

$$\sum_{i=1}^k a_i v_i \quad (a_i \in \text{GF}(q))$$

are precisely the distinct vectors of C .

□

2 Linear codes

The theorems of this section are from [1] and follow the original enumeration of the book.

Note about notation:

A q -ary linear code $[n, k, d]$ -code is also a q -ary (n, q^k, d) -code, but the opposite is not always true.

2.1 Some definitions

Let q be a prime power. View $(F_q)^n$ as the vector space $V(n, q)$.

Definition (Linear code over $GF(q)$). A *linear code over $GF(q)$* is a subspace of $V(n, q)$, with $n \geq 0$.

Thus $C \subseteq V(n, q)$ is a linear code iff

- i. $u + v \in C \quad \forall u, v \in C$
- ii. $au \in C \quad \forall u \in C, a \in GF(q)$

If C is a k -dimensional subspace of $V(n, q)$, then the linear code C is called an $[n, k]$ -code. (sometimes with minimum distance d : $[n, k, d]$ -code).

Definition (Weight). Weight, $w(x)$, of a vector $x \in V(n, q)$ is the number of non-zero entries of x .

Lemma 5.1. if $x, y \in V(n, q)$, then $d(x, y) = w(x - y)$.

Proof. the vector $x - y$ has non-zero entries in precisely those places where x and y differ. $\square$

Theorem 5.2. let C a linear code, let $w(C)$ be the smallest of weights of the non-zero codewords of C . Then $d(C) = w(C)$.

Proof. $\exists x, y \in C$ s.th. $d(C) = d(x, y)$.

By Lemma 5.1, $d(C) = w(x - y) \geq w(C)$, since $x - y$ is a codeword of the linear code C (ie. $x - y \in C$).

On the other hand, for $x \in C$,
 $w(C) = w(x) = d(x, 0) \geq d(C)$, since $0 \in C$.

Hence, $d(C) \geq w(C)$ and $w(C) \geq d(C) \implies d(C) = w(C)$. $\square$

Definition (Generator matrix (GM)). a $k \times n$ matrix whose rows form a basis of a linear $[n, k]$ -code is called a *generator matrix* (GM) of the code.

2.2 Equivalence

Theorem 5.4. two known $k \times n$ matrices generate equivalent linear $[n, k]$ -codes over $GF(q)$ if one matrix can be obtained from the other by a sequence of operations of the following types:

- R1. permutation of the rows
- R2. multiplication of a row by a non-zero scalar
- R3. addition of a scalar multiple of one row to another
- C1. permutation of the columns
- C2. multiplication of any column by a non-zero scalar

Theorem 5.5. let G a GM of an $[n, k]$ -code. Then by performing the previous operations, G can be transformed to the standard form $[I_k | A]$ where I_k is the $k \times k$ identity matrix, and A is a $k \times (n - k)$ matrix.

2.3 Encoding with a linear code

Let C a $[n, k]$ -code over $GF(q)$ with GM G . C contains q^k codewords $\implies$ can be used to communicate any one of q^k distinct messages.

Identify these messages with the q^k tuples of $V(k, q)$.

Encoding:

msg vector: $u = u_1u_2 \dots u_k$; rows of G : $r_1r_2 \dots r_k$. Encode

$$uG = \sum_{i=1}^k u_i r_i \in C$$

ie. uG is a codeword of C ; a linear combination of the rows of the GM.

Encoding function:

$$\begin{aligned} V(k, q) &\longrightarrow C \subseteq V(n, q) \\ u &\longmapsto uG \end{aligned}$$

If G is in standard form (5.5), $G = [I_k | A]$ with $A = [a_{ij}]$, then encoding u is:

$$x = uG = x_1x_2 \dots x_kx_{k+1} \dots x_n$$

where

$$\begin{aligned} x_i &= u_i \quad \text{for } 1 \leq i \leq k, \text{ message digits} \\ x_{k+i} &= \sum_{j=1}^k a_{ji}u_j \quad \text{for } 1 \leq i \leq n-k, \text{ check digits} \end{aligned}$$

the check digits add redundancy to give protection against noise.

2.4 Decoding with a linear code

Codeword $x = x_1x_2 \dots x_n$, received vector $y = y_1y_2 \dots y_n$, error vector $e = y - x = e_1e_2 \dots e_n$.

2.4.1 Cosets

Cosets here are similar to the ones from typical group theory, just from another perspective.

Definition (Coset). Let C a $[n, k]$ -code over $GF(q)$ and $a \in V(n, q)$. Then the set $a + C = \{a + x | x \in C\}$ is called a *coset* of C .

Definition (Coset leader). the vector having minimum weight in a coset is called the *coset leader*.

Lemma 6.3. Let $a + C$ a coset of C and $b \in a + C$. Then $b + C = a + C$.

Proof. since $b \in a + C \implies b = a + x$ for some $x \in C$. If $b + y \in b + C$ then

$$b + y = (a + x) + y = a + (x + y) \in a + C$$

For the other direction, if $a + x \in a + C$ then

$$a + x = (b - x) + z = b + (z - x) \in b + C$$

hence $a + C \subseteq b + C$.

Therefore $b + C = a + C$. $\square$

Theorem 6.4 (Lagrange). suppose $C \subseteq V(n, q)$ is a $[n, k]$ -code over $GF(q)$. Then

- i. every vector of $V(n, q)$ is in some coset of C
- ii. every coset contains exactly q^k vectors
- iii. two cosets either are disjoint or coincide (partial overlap is impossible)

Proof. i. if $a \in V(n, q)$ then $a = a + 0 \in a + C$

ii. the map

$$\begin{aligned} C &\longrightarrow a + C \quad \forall x \in C \\ x &\longmapsto a + x \end{aligned}$$

is a one-to-one map, hence $|a + C| = |C| = q^k$.

- iii. by contradiction, suppose cosets $a + C$, $b + C$ overlap. Then for some vector v ,

$$v \in (a + C) \cap (b + C)$$

thus for some $x, y \in C$, $v = a + x = b + y$.

Hence $b = a + (x - y) \in a + C$, thus by Lemma 6.3 we have $b + C = a + C$. $\square$

Corollary . 6.4 shows that $V(n, q)$ is partitioned into disjoint cosets of C :

$$V(n, q) = (0 + C) \cup (a_1 + C) \cup \dots \cup (a_s + C)$$

where $s = q^{n-k} - 1$, and by Lemma 6.3 we may take $0, a_1, \dots, a_s$ to be coset leaders.

Example 6.5. Let C be the binary $[4, 2]$ -code with generator matrix

$$G = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

ie. $C = \{0000, 1011, 0101, 1110\}$.

Then the cosets of C are

$$\begin{aligned} 0000 + C &= C \text{ itself} \\ 1000 + C &= \{1000, 0011, 1101, 0110\} \\ 0100 + C &= \{0100, 1111, 0001, 1010\} = 0001 + C \\ 0010 + C &= \{0010, 1001, 0111, 1100\} \end{aligned}$$

by Lemma 6.3: $0001 \in 0100 + C$, thus $0001 + C = 0100 + C$.

2.5 Probability of error correction (in binary linear codes)

To evaluate how good a code is, must calculate the *probability* that a received vector will be decoded as the codeword which was sent.

Let p be the symbol error probability.

Probability that the error vector is given vector of weight i is $p^i(1-p)^{n-i}$.

Theorem 6.7. Let C a binary $[n, k]$ -code, and for $i = 0, 1, \dots, n$ let α_i denote the number of coset leaders of weight i .

Then the probability $P_{corr}(C)$ that a received vector decoded by means of a standard array is the codeword which was sent is

$$P_{corr}(C) = \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i}$$

Example 6.8. $[n, k]$ -code from 6.5, coset leaders are 0000, 1000, 0100, 0010. Hence $\alpha_0 = 1$, $\alpha_1 = 3$, $\alpha_2 = \alpha_3 = \alpha_4 = 0$, so

$$P_{corr}(C) = (1-p)^4 + 3p(1-p)^3 = (1-p)^3(1+2p)$$

Lemma 6.9 (Remark 6.9). if $d(C) = 2t + 1$ or $d(C) = 2t + 2$, then C can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

Definition (Rate). $[n, k]$ -code C uses n symbols to send k message symbols; it has *rate* $R(C) = k/n$.

A good code will have a high rate.

Definition (Capacity). the *capacity* $\mathcal{C}(p)$ of a binary symmetric channel with symbol error probability p is

$$\mathcal{C}(p) = 1 + p \cdot \log_2(p) + (1-p) \cdot \log_2(1-p)$$

Theorem 6.12 (Shanon's theorem). channel binary symmetric with symbol error probability p .

Let $R < \mathcal{C}(p)$.

Then for any $\epsilon > 0$, $\exists$ for sufficient large n , an $[n, k]$ -code C of rate $k/n \geq R$ such that $P_{err}(C) < \epsilon$.

Definition (P_{symp}). P_{symp} denotes the average probability that a message symbol is in error after decoding.

Theorem 6.14. let C a binary $[n, k]$ -code and let A_i denote the number of codewords of C of weight i .

Then, if C is used for error detection, the probability of an incorrect message being undetected is

$$P_{\text{undetec}}(C) = \sum_{i=1}^n A_i \cdot p^i (1-p)^{n-i}$$

Example 6.15. for the code of 6.5,

$$P_{\text{undetec}}(C) = \underbrace{1 \cdot p^2 (1-p)^2}_{0101} + \underbrace{2 \cdot p^3 (1-p)^{4-3=1}}_{1011, 1110} = p^2 - p^4$$

2.6 Exercises

Exercise 5.1. is the binary $(11, 24, 5)$ -code linear?

Proof. recall: a q -ary linear $[n, k, d]$ -code is also a q -ary (n, q^k, d) -code, but the opposite is not always true.

$\implies$ $(11, 24, 5)$ -code to be linear it would mean that $24 = q^k$.

Since it is binary, $q = 2$. But $24 \neq 2^k$, so it can not be linear. $\square$

Exercise 5.2. E_n is the code of all even-weight vectors of $V(n, 2)$, which is linear. What are the parameters $[n, k, d]$ of E_n ? Write a GM for E_n in standard form.

Proof.

$$E_n = \{x \in V(n, 2) | w(x) \text{ is even}\}$$

$\longrightarrow$ is the binary even-weight code (the single parity-check code); it's linear since

- the zero vector has even weight
- over $GF(2)$, the sum of two even-weight vectors has even weight

Recall: $F_q^n \iff V(n, q)$, so $V(n, 2) \cong F_2^n$

$$\implies E_n = \{(x_1, \dots, x_n) \in F_2^n | x_1 + \dots + x_n = 0\}$$

minimum distance is 2, since

- no word of weight 1 is in E_n
- words of weight 2 are in E_n

Therefore, parameters are $[n, n-1, 2]$

GM:

$$G = [I_{n-1} | 1]$$

which generates vectors of the form

$$(a_1, \dots, a_{n-1}, a_1 + \dots + a_{n-1})$$

□

Exercise 5.5. Prove that in a *binary linear code*, either all the codewords have even weight or exactly half have even weight and half odd weight.

Proof. let C_e, C_o be subsets of C with even and odd weights respectively.

So that $C = C_e \cup C_o$ (disjoint union).

For binary vectors x, y :

$$w(x+y) = w(x) + w(y) \pmod{2}$$

ie. over F_2 , the parity of the weight is additive:

$$even + even = even$$

$$even + odd = odd$$

$$odd + odd = even$$

Now we have two possible cases:

- C has no odd weighted codewords:
 $C_o = \emptyset, \forall x \in C$ x is even weighted
- C has at least one odd-weighted codeword:
let $u \in C_o$, consider the map

$$\begin{aligned} \phi : C_e &\longrightarrow C_o \\ c &\longmapsto c + u \end{aligned}$$

with $c \in C_e, u \in C_o, \phi(c) \in C_o$.

ϕ is bijective, since $\phi(\phi(c)) = (c + u) + u = c$,

so ϕ is it's own inverse; hence $|C_e| = |C_o|$.

Since $C = C_e \cup C_o$, half codewords are even weighted and half odd.

□

Exercise 6.1. construct standard arrays for codes having each of the GMs:

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad G_2 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$$

Proof. recall: for a binary code with GM G , the code C consists of all binary linear combinations of the rows of G .

For G_1 :

rows: $r_1 = (1, 0)$, $r_2 = (0, 1)$. A codeword is any linear combination:

$$uG_1 = a \cdot r_1 + b \cdot r_2, \text{ with } a, b \in \{0, 1\}$$

4 combinations:

$$0 \cdot r_1 + 0 \cdot r_2 = (0, 0)$$

$$1 \cdot r_1 + 0 \cdot r_2 = (1, 0)$$

$$0 \cdot r_1 + 1 \cdot r_2 = (0, 1)$$

$$1 \cdot r_1 + 1 \cdot r_2 = (1, 0) + (0, 1) = (1, 1)$$

$$\implies C_1 = \{00, 10, 01, 11\}$$

standard array: 00 01 10 11

For G_2 :

rows: $r_1 = (1, 0, 1)$, $r_2 = (0, 1, 1)$. Codeword: $a \cdot r_1 + b \cdot r_2$ with $a, b \in \{0, 1\}$.

4 combinations:

$$0 \cdot r_1 + 0 \cdot r_2 = (0, 0, 0)$$

$$1 \cdot r_1 + 0 \cdot r_2 = (1, 0, 1)$$

$$0 \cdot r_1 + 1 \cdot r_2 = (0, 1, 1)$$

$$1 \cdot r_1 + 1 \cdot r_2 = (1, 0, 1) + (0, 1, 1) = (1, 1, 0)$$

$$\implies C_2 = \{000, 101, 011, 110\}$$

$\implies$ so the 1st row of the standard array is: 000 101 011 110.

Now, pick a non used vector: 001, add it to the code C_2 :

$$001 + C_{2i}$$

$$001 + 000 = 001$$

$$001 + 101 = 100$$

$$001 + 011 = 010$$

$$001 + 110 = 111$$

$\implies$ 2nd row of the standard array: 001 100 010 111

Thus the standard array is

$$000 \ 101 \ 011 \ 110$$

$$001 \ 100 \ 010 \ 111$$

ie. $2^3 = 8$ vectors, a $V(3, 2)$ vector space.

□

Exercise 6.4. C a binary $[n, k]$ -code with minimum distance $2t + 1$ (or $2t + 2$). Given that p is very small, show that an approximate value of $P_{err}(C)$ is $\left(\binom{n}{t+1} - \alpha_{t+1}\right) p^{t+1}$, where α_{t+1} is the number of coset leaders of C of weight $t + 1$.

Proof. From 6.9, if $d(C) = 2t + 1$ (or $2t + 2$) $\Rightarrow C$ can correct any t errors.

Hence every vector of weight $\leq t$ is a coset leader and so $\alpha_i = \binom{n}{i}$ for $0 \leq i \leq t$.

$$P_{corr}(C) = \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i} \quad (1)$$

where α_i is the number of coset leaders of weight i

From the $\binom{n}{i}$ vectors of weight i , exactly α_i are coset leaders, so the non coset leaders are $\binom{n}{i} - \alpha_i$. This is when a decoding error happens.

$$\Rightarrow P_{err}(C) = \sum_{i=0}^n \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i}$$

Alternatively from 1:

$$P_{err}(C) = 1 - P_{corr}(C) = 1 - \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i}$$

Notice that

$$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} = (p + (1-p))^n = 1^n = 1$$

thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} - \sum_{i=0}^n \alpha_i \cdot p^i (1-p)^{n-i} \\ &= \sum_{i=0}^n \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \end{aligned}$$

Now, from 6.9, $\forall i \leq t$, $\alpha_i = \binom{n}{i}$, thus $\binom{n}{i} - \alpha_i = 0$ for the first t terms in the sum.

Thus

$$\begin{aligned} P_{err}(C) &= \sum_{i=0}^n \left(\binom{n}{i} - \alpha_i \right) p^i (1-p)^{n-i} \\ &= \left(\binom{n}{t+1} - \alpha_{t+1} \right) p^{t+1} (1-p)^{n-t-1} + \dots \end{aligned}$$

where the ... represent the terms with p^{t+2} and higher.

Then, since p is very small (by the exercise definition), $(1 - p)^{n-t+1} \equiv 1$, and the other higher powers of p are negligible.

Hence,

$$P_{err}(C) = \left(\binom{n}{t+1} - \alpha_{t+1} \right) \cdot p^{t+1}$$

□

3 Dual code, PCM and Syndrome

Definition (Orthogonal). Inner product:

$$u \cdot v = u_1 v_1 + \dots + u_n \cdot v_n = \sum_{i=1}^n u_i v_i$$

If $u \cdot v = 0$, then u and v are called *orthogonal*.

Lemma 7.1. for any $u, v, w \in V(n, q)$ and $\lambda, \mu \in GF(q)$,

- i. $u \cdot v = v \cdot u$
- ii. $(\lambda u + \mu v) \cdot w = \lambda(u \cdot w) + \mu(v \cdot w)$

Proof. (at exercise 7.1). □

Definition (Dual code). given a linear $[n, k]$ -code C , the *dual code* of C , denoted $C^\perp$, is defined to be the set of those vectors of $V(n, q)$ which are orthogonal to every codeword of C . ie.

$$C^\perp = \{v \in V(n, q) \mid v \cdot u = 0 \forall u \in C\}$$

Lemma 7.2. C a $[n, k]$ -code with GM G . Then $v \in V(n, q)$ belongs to $C^\perp$ iff v is orthogonal to every row of G , ie.

$$v \in C^\perp \iff v \cdot G^T = 0$$

Proof. let rows of G be $r_1, \dots, r_k$, suppose $v \cdot r_i = 0 \forall i$ (ie. v orthogonal to every row of G).

If $u \in C$ (ie. a codeword of C), then $u = \sum_{i=1}^k \lambda_i r_i$ for some scalars λ_i , so that

$$v \cdot u = v \cdot \sum_{i=1}^k \lambda_i r_i = \sum_{i=1}^k \lambda_i \cdot \underbrace{(v \cdot r_i)}_0 = \sum \lambda_i \cdot 0 = 0$$

$\implies$ hence v is orthogonal to every codeword of C , and so is in $C^\perp$. □

Theorem 7.3. C a $[n, k]$ -code over $GF(q)$. Then the dual code $C^\perp$ of C is a linear $[n, n - k]$ -code.

Proof. i. first show that $C^\perp$ is a linear code:

let $v_1, v_2 \in C^\perp$ and $\lambda, \mu \in GF(q)$. Then $\forall u \in C$,

$$\begin{aligned} (\lambda v_1 + \mu v_2) \cdot u &= \lambda(v_1 \cdot u) + \mu(v_2 \cdot u) \\ &\text{notice that } v_1, v_2 \in C^\perp, \text{ so} \\ &\text{that } v_1, v_2 \text{ are orthogonal with } u \\ &\text{hence } (v_1 \cdot u) \text{ and } (v_2 \cdot u) \text{ are } 0 \\ &= \lambda \cdot 0 + \mu \cdot 0 = 0 \end{aligned}$$

Hence $\lambda \cdot v_1 + \mu \cdot v_2 \in C^\perp$, and so $C^\perp$ is linear, since it is closed under linear combinations.

ii. Next, show that $C^\perp$ has dimension $n - k$:

Let $G = [g_{ij}]$ be a GM of C .

Then by Lemma 7.2, the elements of $C^\perp$ are the vectors $v = (v_1, v_2, \dots, v_n)$, since $v \in C^\perp \iff v \cdot G^T = 0$, satisfying

$$\sum_{j=1}^m g_{ij} v_j = 0 \quad \forall i = [k]$$

□

Exercise 7.1. (wip)

References

- [1] Raymond Hill. A First Course in Coding Theory, 1986.
- [2] J.H. van Lint. Introduction to Coding Theory, 1981.